Ridge Regression on Diamonds Dataset

Comparative Analysis of Optimization Algorithms

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Introduction

The primary objective of this study is to conduct a comparative analysis of the convergence behavior of various optimization algorithms applied on a Ridge Regression problem to the Diamonds Dataset.

In particular, the focus is on:

- Evaluating the respective algorithms convergence rates.
- Investigating how theoretical properties of the objective function (such as Smoothness and Strongly Convexity) have an impact to results.
- Evaluating how the L_2 regularization parameter (λ) of Ridge Regression influences speed and efficiency of the algorithms.

The following algorithms were implemented:

- Gradient Descent (Naive, Bounded, Smooth).
- Accelerated Gradient Descent (Nesterov's).
- Stochastic Gradient Descent (Naive, Strongly Convex)
- Newton's Method.
- Adagrad.

Dataset Overview: The Diamonds Dataset

To evaluate the performance of the optimization algorithms, we utilized the **Diamonds Dataset**. The dataset consists of approximately **54,000 observations** ($n = 53,940$), each representing a diamond with **10 distinct attributes**.
The primary objective is to predict with a Ridge Regression the **Price** of a diamond based on its physical and quality characteristics.

Feature Description:

- **Target Variable (y):** `price` in US dollars.
- **Numerical Features:**
  - `carat`: Weight, the most significant predictor.
  - `x`, `y`, `z`: Length, Width, and Depth in mm.
  - `depth`: Total depth percentage ($z/\text{mean}(x, y)$).
  - `table`: Width of the top relative to the widest point.
- **Categorical Features (Ordinal):** `cut`, `color`, and `clarity`.

Dataset Overview: The Diamonds Dataset

Distribution of all the main variables within the diamonds dataset.

Dataset Characteristics: Multicollinearity

What is Multicollinearity?
It occurs when two or more independent variables in a regression model are highly linearly correlated, meaning one feature can be accurately predicted from the others.

The Diamonds Case:
In our specific dataset, there is a strong correlation between `carat` (weight) and the physical dimensions (`x`, `y`, `z`). These features provide almost identical information.

Impact on Optimization:

- This structural redundancy results in a feature matrix X with a very small minimum eigenvalue ($\lambda_{\text{min}} \approx 0$).
- Consequently, the underlying optimization problem becomes highly **ill-conditioned**. It is particularly challenging for standard gradient-based methods because it creates flat, unstable valleys in the loss landscape.

Ridge Regression

In this study, we address the **Ridge Regression** problem, which extends Ordinary Least Squares (OLS) by imposing an L_2 penalty on the model parameters to mitigate overfitting and multicollinearity.

The objective function is the sum of squared errors for each data point ($i = 1$ to n) plus the sum of squared weights (from $j = 1$ to d).

$$f(w) = \frac{1}{2n} \sum_{i=1}^{n} (y_i - x_i^T w)^2 + \frac{\lambda}{2} \sum_{j=1}^{d} w_j^2$$

The objective function $f(w)$ is defined in vector notation as follows.

$$f(w) = \underbrace{\frac{1}{2n} \|y - Xw\|^2}_{\text{Half MSE}} + \underbrace{\frac{\lambda}{2} \|w\|^2}_{\text{Regularization}}$$

Where:

- $X \in \mathbb{R}^{n \times d}$ is the feature matrix (standardized, includes a column of 1).
- $y \in \mathbb{R}^{n}$ is the target vector (price).
- $w \in \mathbb{R}^{d}$ is the vector of weights to be optimized.
- $\lambda > 0$ is the regularization hyperparameter.

Ridge Regression: Why the scaling?

Why the scaling?
Introducing specific scaling factors is convenient for mathematical and numerical stability:

- **The factor $\frac{1}{2}$**: Cancels out the exponent 2 when computing the gradient $\nabla f(w)$ and the derivative of $\|w\|^2$, simplifying to a linear form without constant multipliers.
- **The factor $\frac{1}{n}$**: Normalizes the error term with respect to the dataset size. This ensures that the magnitude of the loss (and consequently the gradient) is independent of the number of samples n. This allows the same learning rates to be effective regardless of whether we use the full dataset or a smaller subset.

Gradient and Hessian Matrix

Since $f(w)$ is differentiable, we can work with first-order derivative (the Gradient) and second-order derivative (the Hessian matrix) to minimize the objective function.

The Gradient $\nabla f(w)$
The gradient represents the steepest ascent direction of the function. To find the minimum, optimization algorithms like Gradient Descent move in the opposite direction of the gradient.
Let's expand the vector notation of the objective function:

$$f(w) = \frac{1}{2n}\|y - Xw\|^2 + \frac{\lambda}{2}\|w\|^2$$

$$f(w) = \frac{1}{2n}(y - Xw)^T(y - Xw) + \frac{\lambda}{2}w^T w$$

The Gradient $\nabla f(w)$

Let's expand the error quadratic term:

$$\begin{aligned}(y - Xw)^T(y - Xw) &= (y^T - (Xw)^T)(y - Xw) \\ &= (y^T - w^TX^T)(y - Xw) \\ &= y^Ty - y^TXw - w^TX^Ty + w^TX^TXw\end{aligned}$$

The terms y^TXw and w^TX^Ty are scalars (yielding a 1×1 result). In linear algebra, the transpose of a scalar is the scalar itself. So $(y^TXw)^T = w^TX^Ty$. Therefore, these two terms are identical and can be combined:

$$-y^TXw - w^TX^Ty = -2w^TX^Ty$$

Now we substitute all in the original function:

$$f(w) = \frac{1}{2n}\left(y^Ty - 2w^TX^Ty + w^TX^TXw\right) + \frac{\lambda}{2}w^Tw$$

The Gradient $\nabla f(w)$

The gradient is the vector of partial derivatives with respect to w. To compute the gradients and Hessians, we rely on the following standard matrix calculus properties (assuming w and b are column vectors, and A is a symmetric matrix):

1. **Constant rule:** $\nabla_w(b) = 0$
2. **Linear form:** $\nabla_w(w^T b) = \nabla_w(b^T w) = b$
3. **Jacobian of a linear mapping:** $\nabla_w(Aw) = A$
4. **Quadratic form:** $\nabla_w(w^T Aw) = 2Aw$ *(Requires A to be symmetric, i.e., $A = A^T$. In our case, $X^T X$ and I are both symmetric).*

Let's apply these rules to each term of $f(w)$:

1. **Term** $y^T y$: Does not depend on w.

$$\nabla_w(y^T y) = 0$$

2. **Term** $-2w^T(X^T y)$: Here $b = X^T y$.

$$\nabla_w\left(\frac{1}{2n}(-2w^T X^T y)\right) = -\frac{1}{n}\nabla_w(w^T(X^T y)) = -\frac{1}{n}X^T y$$

The Gradient $\nabla f(w)$

3. **Term** $w^T(X^TX)w$: Here $A = X^TX$. Note that X^TX is symmetric.

$$\nabla_w \left(\frac{1}{2n} w^T X^T X w \right) = \frac{1}{2n}(2X^TXw) = \frac{1}{n}X^TXw$$

4. **Regularization term:** $\frac{\lambda}{2}w^Tw$: Here $A = I$ (identity matrix).

$$\nabla_w \left(\frac{\lambda}{2} w^T I w \right) = \frac{\lambda}{2}(2Iw) = \lambda w$$

Gradient Result:
Summing all the parts together:

$$\nabla f(w) = \frac{1}{n}(X^TXw - X^Ty) + \lambda w$$

Factoring out common terms, this is also frequently written as:

$$\nabla f(w) = \frac{1}{n}X^T(Xw - y) + \lambda w$$

Gradient is a $d \times 1$ vector.

The Hessian Matrix $\nabla^2 f(w)$

The Hessian matrix is the derivative of the gradient with respect to w (i.e., the second derivative of the original function). It is a $d \times d$ matrix.
Let's recall the expression for the gradient:

$$\nabla f(w) = \frac{1}{n}X^T Xw - \frac{1}{n}X^T y + \lambda w$$

We differentiate each term with respect to w. We use the matrix calculus identity $\nabla_w(Aw) = A$:

1. **Term** $\frac{1}{n}X^T Xw$: The constant matrix multiplying w is $\frac{1}{n}X^T X$.

$$\nabla_w \left(\frac{1}{n}X^T Xw\right) = \frac{1}{n}X^T X$$

The Hessian Matrix $\nabla^2 f(w)$

2. **Term** $-\frac{1}{n}X^T y$: This term does not contain w (it is a constant vector).

$$\nabla_w \left(-\frac{1}{n}X^T y\right) = 0$$

3. **Term** λw: We can rewrite this as $\lambda I w$. The constant matrix is λI.

$$\nabla_w(\lambda I w) = \lambda I$$

We put the together, so the **Hessian Matrix** is :

$$H = \nabla^2 f(w) = \frac{1}{n}X^T X + \lambda I$$

$f(w)$ is strictly convex

To investigate the convexity of our objective function, we rely on the formal definition of definite matrices based on their quadratic forms.

A symmetric matrix M is Positive Semi-Definite if $v^T M v \geq 0$ for all vectors v, and Positive Definite if $v^T M v > 0$ for all non-zero vectors $v \neq 0$.

Let's apply this definition to our Ridge Regression Hessian matrix, $H = \frac{1}{n} X^T X + \lambda I$, by analyzing its quadratic form $v^T H v$ for any arbitrary non-zero vector $v \in \mathbb{R}^d$:

$$v^T H v = v^T \left(\frac{1}{n} X^T X + \lambda I \right) v$$

By distributing the multiplication, we get:

$$v^T H v = \frac{1}{n} v^T X^T X v + \lambda v^T I v$$

$f(w)$ is strictly convex

We can rewrite the first term using the property of transposes, $(Xv)^T = v^T X^T$, and the second term using the definition of the squared L_2 norm, $v^T v = \|v\|_2^2$:

$$v^T H v = \frac{1}{n}(Xv)^T(Xv) + \lambda\|v\|_2^2$$

$$v^T H v = \underbrace{\frac{1}{n}\|Xv\|^2}_{\text{OLS Base}} + \underbrace{\lambda\|v\|^2}_{\text{Ridge Penalty}}$$

Now, let's evaluate the two components:

- **The OLS Base:** The squared norm $\|Xv\|^2$ is a squared length, meaning it is always non-negative (≥ 0). This proves that the OLS Hessian ($\frac{1}{n}X^T X$) is intrinsically **Positive Semi-Definite**. However, if features are perfectly collinear, there exist non-zero vectors v where $Xv = 0$. In these cases, the quadratic form equals zero, meaning the loss function has flat valleys with no unique minimum.

$f(w)$ is strictly convex

- **The Ridge Penalty:** For any non-zero vector $v \neq 0$, its squared norm $\|v\|^2$ is strictly positive. Since our regularization hyperparameter is strictly positive ($\lambda > 0$), the entire term $\lambda\|v\|^2$ must be strictly greater than zero (> 0).

Because we are adding a strictly positive term ($\lambda\|v\|^2 > 0$) to a non-negative term ($\frac{1}{n}\|Xv\|^2 \geq 0$), the entire sum is guaranteed to be strictly positive:

$$v^T H v > 0 \quad \text{for all } v \neq 0$$

By the mathematical definition, this proves that the Ridge Hessian matrix H is **Positive Definite**. The L_2 penalty mathematically eliminates any flat regions, guaranteeing that the objective function possesses a single, unique global minimum, so that means $f(w)$ is strictly convex.

Ridge Regression - Properties

To understand how iterative optimization algorithms will behave on Ridge Regression objective function $f(w)$, we must analyze its properties.

1. Convexity: Yes (Strictly Convex)
As proven in the previous section using the Hessian's quadratic form, the addition of the L_2 penalty ensures that the Hessian matrix H is Positive Definite. Therefore, the function is not just convex, but **strictly convex**. This guarantees that any local minimum is the unique global minimum w^*.

2. Lipschitz Continuous: No (Globally)
A function $f(w)$ is globally B-Lipschitz if its gradients are bounded in magnitude everywhere ($\|\nabla f(w)\| \leq B$). Our objective function $f(w)$ is a quadratic function. As the weights w can grow toward infinity, the gradient $\nabla f(w) = \frac{1}{n}X^T(Xw - y) + \lambda w$ also can grows linearly toward infinity. Because the gradient is unbounded over the entire domain $\mathbb{R}^d$, the function itself is **not globally Lipschitz continuous**.

Ridge Regression - Properties

3. Smoothness (L-smooth): Yes

It is L-**Smooth**. The curvature does not change abruptly; the gradient doesn't oscillate wildly. The function always lies *below* a certain parabola:

$$f(y) \leq f(x) + \nabla f(w)^T (y - x) + \frac{L}{2}\|x - y\|^2$$

- **Value of L:** It corresponds to the largest eigenvalue of the Hessian matrix (the maximum curvature of the loss landscape).

$$L = \lambda_{\max}\left(\frac{1}{n}X^T X\right) + \lambda$$

- **Consequence:** This value acts as the absolute **speed limit** for Gradient Descent. The optimal safe step size is typically $\gamma = \frac{1}{L}$.

Ridge Regression - Properties

4. Strong Convexity (μ-strongly convex): Yes
It is μ-**Strongly Convex**. The curvature is "at least" μ in every direction. The function always lies *above* the quadratic function:

$$f(y) \geq f(x) + \nabla f(x)^T (y - w) + \frac{\mu}{2} \|x - y\|^2$$

- **Value of μ:** It corresponds to the smallest eigenvalue of the Hessian matrix.

$$\mu = \lambda_{\min} \left(\frac{1}{n} X^T X \right) + \lambda$$

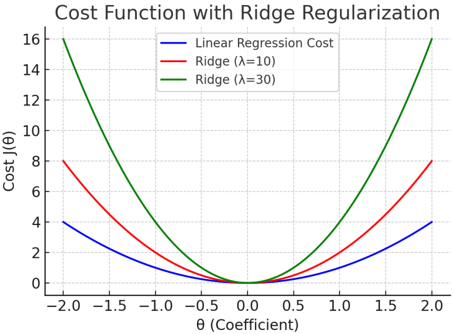
If the feature matrix X is not full-rank, for instance, due to perfectly collinear features, the minimum eigenvalue of the OLS component $\frac{1}{n} X^T X$ is exactly 0, meaning $\mu = \lambda$. This mathematically proves that without regularization ($\lambda = 0$), the problem might not be strongly convex, which would make convergence much slower.

- **Consequence:** The larger μ is, faster the algorithm will converge to the minimum.

The expected effect of L2 regularization

We expect the regularization parameter λ alter the geometry of our cost function.

Specifically, increasing λ directly increases μ, the **strong convexity parameter**, which dictates the *minimum curvature* of the loss landscape. Mathematically, by adding the quadratic penalty $\frac{\lambda}{2}\|w\|_2^2$, we are guaranteeing that the function curves upwards by at least a factor of λ in every single direction. Visually, this means any "flat valleys" caused by multicollinearity are physically eliminated, making the entire bowl narrower and more symmetrically curved .



The expected effect of L2 regularization

Expected Results on Convergence:

Because the minimum curvature (μ) is strictly bounded away from zero and increases with λ, the loss function no longer has any flat regions where the algorithm might stall.

The gradients will always provide a strong and clear signal pointing toward the minimum. Consequently, we expect that as λ increases, the Gradient Descent algorithm will be pushed toward the global minimum much more rapidly, requiring significantly fewer iterations to reach the strict convergence criterion.

Dataset pre-processing steps

To prepare the dataset for our optimization algorithms, we now apply the following preprocessing steps:

1. **Ordinal Encoding:** Categorical variables (`cut`, `color`, `clarity`) were mapped to integer values, preserving their inherent hierarchy. This safely converts text labels into a numerical format that the mathematical model can process.

2. **Removing Outliers:** We filter out extreme anomalous data points. Since our objective function relies on Mean Squared Error (MSE), it is highly sensitive to outliers, which could disproportionately skew the optimization of our weights.

Before feeding our data into the optimization algorithm, we must carefully preprocess our matrices.

Dataset pre-processing steps

1. **Standardizing the Features (X) and the Target (y):** By subtracting the mean and dividing by the standard deviation, we ensure all features and the target share the same scale (a mean of 0 and a variance of 1). Mathematically, for each feature column x_j in X and for the target vector y, the transformation is defined as:

$$x_j = \frac{x_j - \mu_j}{\sigma_j} \qquad y = \frac{y - \mu_y}{\sigma_y}$$

   This is crucial because it prevents features with naturally larger ranges from dominating the cost function. Geometrically, it transforms the loss landscape from an elongated ellipse into a much more spherical shape, which allows Gradient Descent to point much more directly toward the minimum and converge significantly faster.

2. **The Bias Trick (Adding a column of 1s):** We append a column composed entirely of ones to our feature matrix X. This mathematical trick allows us to absorb the intercept (or bias) term directly into our weight vector w.

Experimental Approach and Goals

We will now test and compare the various optimization algorithms. Each algorithm will be evaluated across a predefined set of regularization hyperparameters: $\lambda \in \{10^{-5}, 10^{-4}, 10^{-3}, 0.01, 0.1, 1.0\}$.

The objective of this experiment is to conduct comparative analysis with a focus on:

- **Impact of λ on convergence:** On the same algorithm, how varying the λ penalty impacts the number of iterations and time required to reach convergence.
- **Compare different algorithms** How different algorithms(GD, SGD, Newton methods ecc..) performs under the same regularization conditions.

Exact Analytical Solution

We compute the exact global minimum w^* and its corresponding optimal loss $f(w^*)$ across the predefined set of regularization hyperparameters $\lambda \in \{10^{-5}, 10^{-4}, 10^{-3}, 0.01, 0.1, 1.0\}$. By knowing the absolute minimum w^* beforehand, we can use it later as a benchmark to evaluate exactly how well and how fast our iterative algorithms converge to the true solution.

To find this baseline, we directly apply the **closed-form solution** using `np.linalg.solve(A, b)` which efficiently and stably solves the underlying linear system of equations

$$\left(\frac{1}{n}X^T X + \lambda I\right) w = \frac{1}{n}X^T y$$

providing us with a accurate weight vector so we can compute the optimal values for each λ.

Exact Analytical Solution

λ	10^{-5}	10^{-4}	10^{-3}	0.01	0.1	1.0
$f(w^*)$	0.04265771	0.04276069	0.04369175	0.04977553	0.07101217	0.16355421

As the regularization hyperparameter λ increases, the training error (loss) increases as well. This happens because the optimization algorithm is no longer solely focused on minimizing the residual errors to fit the training data perfectly.

Instead, it must balance that goal with keeping the model weights (w) as small as possible to minimize the heavy L_2 penalty. By forcing the weights to shrink, we restrict the model's flexibility, preventing it from capturing every minor fluctuation and noise in the training set.

Gradient Descent

Now we implement the core iterative optimization algorithm: **Gradient Descent**.

This function updates our weight vector w step-by-step by moving in the opposite direction of the gradient, scaled by the learning rate (step size) γ:

$$w^{(t+1)} = w^{(t)} - \gamma \nabla f(w^{(t)})$$

We have introduce a **stopping criterion**. Rather than running for `max_iters`, the algorithm continuously monitors ϵ (epsilon): the difference between the current loss and the exact analytical minimum w^* we established in the previous step.

If this difference drops below a microscopic tolerance threshold (`tol = 1e-6`), the algorithm recognizes that it has practically reached the global minimum and triggers an "early stop".

$$\epsilon = f(w_T) - f(w^*)$$

Gradient Descent - Naive Learning Rate

Now we execute our Gradient Descent algorithm across our predefined set of L2 λ values.

For this first run, we are intentionally using a **naive, very small learning rate** γ **= 0.0001** .

Choosing a tiny step size is a conservative and highly stable approach: it guarantees that our algorithm will not overshoot the minimum or diverge. However, this safety comes with a heavy trade-off in efficiency. Taking such microscopic steps means the algorithm might require a massive number of iterations to finally reach the bottom of the convex bowl, which is why we must allow a high computational budget (`max_iters = 50000`).

The code loops through each regularization strength, times the execution to measure computational efficiency, and stores all the final metrics—including the optimized weights, final loss, iterations needed to trigger the early stop, and elapsed time.

Gradient Descent - Naive Learning Rate

Final results per λ using a naive learning rate ($\gamma = 0.0001$):

λ	**Loss**	γ	**Iters**	**Time[s]**
10^{-5}	0.05510	0.0001	50000	9.55
10^{-4}	0.05512	0.0001	50000	10.53
10^{-3}	0.05528	0.0001	50000	9.70
0.01	0.05689	0.0001	50000	10.24
0.1	0.07171	0.0001	50000	11.27
1.0	0.16356	0.0001	31209	6.40

Observations:

- For low regularizations (from $\lambda = 10^{-5}$ to 0.1), the algorithm hits the `max_iters` limit of 50,000 without reaching the tight tolerance of the early stop.
- Only for the strongest regularization ($\lambda = 1.0$), the algorithm is pushed fast enough to converge in exactly 31,209 iterations.

Gradient Descent - Assuming Bounded Gradient

Assume we are moving in a bounded region $\|w\| \leq R$ containing all iterates, and the distance between the initial solution and optimal solution is bounded by R. Since we initialize our initial solution vector to **Zero** (`np.zeros`), the distance coincides exactly with the norm of the solution:

$$\|w_0 - w^*\| = \|w^*\| \leq R$$

$$\|0 - w^*\| = \|w^*\| \leq R$$

Basically the assumption on R means: *"The optimal solution and the optimization path to reach it lie within a sphere of radius R."*

- Since we have **standardized the data** (X and y have zero mean and unit variance), the optimized weights w will naturally be small (typically between -1 and 1)
- We set $R = 5.0$. A generous safety margin (a "Bounded Region") that is virtually guaranteed to contain the true solution.

Gradient Descent - Assuming Bounded Gradient

We need to find a number B such that $\|\nabla f(w)\| \leq B$ for every w within the radius R. We rewrite the gradient formula in this way:

$$\nabla f(w) = \left(\frac{1}{n}X^T X + \lambda I\right) w - \frac{1}{n}X^T y$$

Using the triangle inequality ($\|a + (-b)\| \leq \|a\| + \|-b\|$) and the definition of matrix norms:

$$\|\nabla f(w)\| \leq \underbrace{\left\|\frac{1}{n}X^T X + \lambda I\right\|}_{\text{This is smooth parameter L}} \cdot \underbrace{\|w\|}_{\leq R} + \underbrace{\left\|\frac{1}{n}X^T y\right\|}_{\text{Constant}}$$

So the formula for B is:

$$B = L \cdot R + \left\|\frac{1}{n}X^T y\right\|$$

Gradient Descent - Assuming Bounded Gradient

By making the assumption that our optimization path remains within a bounded region $\|w\| \leq R$, and having calculated the maximum gradient bound B such that $\|\nabla f(w)\| \leq B$, we have effectively established that our objective function is B-**Lipschitz continuous** within this domain.

For Lipschitz continuous functions, optimization theory provides a specific, optimal step size: we can now set γ using the formula:

$$\gamma = \frac{R}{B\sqrt{T}}$$

Gradient Descent - Assuming Bounded Gradient

Final results per λ using the bounded gradient step size (γ):

λ	**Loss**	γ	**Iters**	**Time[s]**
10^{-5}	0.04966	0.00214	10000	1.86
10^{-4}	0.04969	0.00214	10000	2.04
10^{-3}	0.04997	0.00214	10000	2.88
0.01	0.05263	0.00214	10000	2.15
0.1	0.07103	0.00210	10000	2.02
1.0	0.16356	0.00176	1768	0.32

Observations:

- **Larger Steps:** The dynamic $\gamma \approx 0.002$ is safely 20x larger than the naive guess.
- **Massive Speedup:** For $\lambda = 1.0$, it converges in only 1,768 iterations (vs. 31,209).
- **Higher Accuracy:** It reaches a strictly lower loss in just 10,000 steps compared to the 50,000 steps of the naive approach.

Gradient Descent - Smoothness

Previously we have seen that function is L-**Smooth**. The curvature does not change abruptly; the gradient does not fluctuate wildly. The function always stays "below" a certain bounding parabola.

- **Value of L:** It is the largest eigenvalue of the Hessian matrix (the maximum curvature).

$$L = \lambda_{\max}\left(\frac{1}{n}X^T X\right) + \lambda$$

or equivalently

$$L = \left\|\frac{1}{n}X^T X\right\| + \lambda$$

- **Consequence:** This value is the **speed limit** for Gradient Descent. If we use a learning rate $\gamma > \frac{2}{L}$, the algorithm will diverge. The optimal safe step size is $1/L$.

Gradient Descent - Smoothness

Final results per λ using the optimal step size ($\gamma = 1/L$):

λ	**Loss**	$\gamma = 1/L$	**Iters**	**Time[s]**
10^{-5}	0.04266	0.23247	8353	1.56
10^{-4}	0.04276	0.23247	7767	1.74
10^{-3}	0.04369	0.23242	4438	1.76
0.01	0.04978	0.23193	765	0.22
0.1	0.07101	0.22719	142	0.07
1.0	0.16355	0.18862	15	0.01

Observations:

- **Optimal Steps:** The safe learning rate ($\gamma \approx 0.23$) is ∼2,300x larger than our naive guess.
- **Lightning Fast:** For $\lambda = 1.0$, it perfectly converges in just 15 iterations (down from 31,209).
- **Perfect Accuracy:** Final loss values now exactly match our analytical baseline.